Tribhuvan University

Faculty of Management

OFFICE OF THE DEAN

2021

Bachelor in Business Administration

Course Title: **Financial Derivatives**

Code No: **BNK202** Semester: **8th Semester**

Full Marks: **60** Pass Marks: **24** Time: **value of both options.d. Assuming options are European, find the lower bound of both options.13.The stock price of Surya Hydro Company is Rs 140. A call option on this stock has exercise price of Rs 125 and selling for Rs 20. Another call option on this stock has Rs 130 exercise price and selling for Rs 12. Both calls are American and expire after 6 months. Explain whether you confirm the rules regarding American calls that differ only by exercise price. Also, suggest arbitrage strategy, if any.14.The stock of NMB bank sells for Rs 500 per share. There exists options on NMB stock, which permit its holder to buy a share at an exercise price of Rs 400. These options will expire after one year at which time stock will be selling either at Rs 800 or at Rs 200. Assume the risk free rate is 8 percent per annum.a. Find the value of call option.b. Show that how could you earn arbitrage profit if market price of call is Rs 220.15.Kathmandu Laboratories, Inc., is a new high-technology company whose common stock sells for Rs 230 per share. A call option exists on this stock with 3 months to expiration. It has an exercise price of Rs 215. You have made a careful study of the stock's volatility and conclude that a standard deviation of 0.50 is appropriate for the next 3 months. Currently, the risk-free interest rate is 6 percent per annum. Using the Black-Scholes-Merton option-pricing model, what is the fair price of European call on this stick?16.Currently the price of Chandragiri Hills stock is 2,500. You want to invest on this stock but you are not sure that whether the price of stock will increase or it will decrease in 6 months. Therefore, you want to make protective put. The put options with Rs 2,500 exercise price on Chandragiri stock are currently selling for Rs 500.a. How can you make protective put?b. What is the cost of protective put to you?c. At what stock price in 6 month will you breakeven?d. Calculate the gain/loss from protective put in 6 months if stock price of Chandragiri at that time turned to be Rs 1,000, Rs 1,500, Rs 2,000, Rs 2,500, Rs 3,000, Rs 3,500, Rs 4,000, or Rs 4,500 respectively and graph the results.Section "C"Comprehensive answer questions:[2 × 10 = 20]17.A pension fund wants to enter into a six-month equity swap with a notional principal of Rs 60 million. Payments will occur in 90 and 180 days. The swap will allow the fund to receive the return on index 1, currently at 5514.67. The fund is considering three different types of swaps, one of which would require it to pay a fixed rate, another that would require it to pay floating rate, and another that would require it to pay the return on index 2, which is currently at 1212.98. Refer to these as swap 1, 2, and 3. The term structure is as follows:TermRateDiscount bond price90 days9%B(90j= 1/(1+.09(90/360)) = 0.9780180 days10%B(0(180)= 1/(1+.10(180/360)) = 0.9524a. Find the fixed rate for swap 1.b. Find the payments on day 90 for swaps 1, 2 and 3. Assume that on day 90 stock index 1 is at 5609.81 and stock index 2 is at 1231.94. Be sure to indicate the net payment.c. Assume it is 30 days into the life of swap. Stock index 1 is at 5499.62 and stock index 2 is at 1201.45. The new term structure is as follow:TermRateDiscount bond price60 days6.80%B30(60)= 1/(1+.068(60/360)) = 0.9888150 days7.05%B30(150)= 1/(1+.0705(150/360)) = 0.9715Find the value of swap 1, 2 and 3.18.A forward contract on silver has one year to expiration. The spot price of silver today is Rs 1,500 per Tola. It requires annual storage cost of Rs 50 per Tola payable in two installments after each six months. Interest rate is currently 8 percent per annum. One contract on silver covers 100 Tola.a. Find the forward price of silver today.b. Show arbitrage strategy if market price of forward contract is Rs 1,625.c. What is the initial value of forward contract?d. Assume it is 3 months after the contract was entered into. The spot price of silver now is Rs 1,350 per Tola and risk-free rate now is 8 percent per annum. What are the forward price and value of short forward contract now?**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions:[10 × 1 =10]

1. **1.** Financial derivatives are purely speculative, and highly leveraged instruments.

2. If you buy a put option while holding the underlying stock, your maximum gain equals the stock purchase price minus the put premium minus the option striking price.

3. Investors who do not consider risk in their decisions are said to be short selling.

4. Selling a call option is a bearish strategy that has limited gain and unlimited loss.

5. There is inverse relation between interest rate and value of call.

6. An option has an exercise price of Rs 25, the company declares a two for one stock split. Adjusted stock price would be Rs 50.

7. Put delta are negative, indicating the facts that the put option price and the underlying stock price are inversely related.

8. In case of certainty, future price is simply the sum of spot price and cost of carry.

9. Swaps are custom-tailored to the need of the counterparties and often sold in organized exchange.

10. In a currency swap, the notional principals are generally exchanged at the beginning and end of swap period.

Section B

Short Answer Questions:[6 × 5 =30]

11. Discuss the current issues of derivative market in Nepal.

12. Consider the following option pricesOptionsCallPutStrike price100100Stock price120105Option price263.5Risk-free interest rate is 6 percent per annum. Options expire in 90 days.a. Which option is in-the-money and which is out-of-the-money?b. Find the intrinsic value of options.c. Find the time value of both options.d. Assuming options are European, find the lower bound of both options.

13. The stock price of Surya Hydro Company is Rs 140. A call option on this stock has exercise price of Rs 125 and selling for Rs 20. Another call option on this stock has Rs 130 exercise price and selling for Rs 12. Both calls are American and expire after 6 months. Explain whether you confirm the rules regarding American calls that differ only by exercise price. Also, suggest arbitrage strategy, if any.

14. The stock of NMB bank sells for Rs 500 per share. There exists options on NMB stock, which permit its holder to buy a share at an exercise price of Rs 400. These options will expire after one year at which time stock will be selling either at Rs 800 or at Rs 200. Assume the risk free rate is 8 percent per annum.a. Find the value of call option.b. Show that how could you earn arbitrage profit if market price of call is Rs 220.

15. Kathmandu Laboratories, Inc., is a new high-technology company whose common stock sells for Rs 230 per share. A call option exists on this stock with 3 months to expiration. It has an exercise price of Rs 215. You have made a careful study of the stock's volatility and conclude that a standard deviation of 0.50 is appropriate for the next 3 months. Currently, the risk-free interest rate is 6 percent per annum. Using the Black-Scholes-Merton option-pricing model, what is the fair price of European call on this stick?

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17. A pension fund wants to enter into a six-month equity swap with a notional principal of Rs 60 million. Payments will occur in 90 and 180 days. The swap will allow the fund to receive the return on index 1, currently at 5514.67. The fund is considering three different types of swaps, one of which would require it to pay a fixed rate, another that would require it to pay floating rate, and another that would require it to pay the return on index 2, which is currently at 1212.98. Refer to these as swap 1, 2, and 3. The term structure is as follows:TermRateDiscount bond price90 days9%B(90j= 1/(1+.09(90/360)) = 0.9780180 days10%B(0(180)= 1/(1+.10(180/360)) = 0.9524a. Find the fixed rate for swap 1.b. Find the payments on day 90 for swaps 1, 2 and 3. Assume that on day 90 stock index 1 is at 5609.81 and stock index 2 is at 1231.94. Be sure to indicate the net payment.c. Assume it is 30 days into the life of swap. Stock index 1 is at 5499.62 and stock index 2 is at 1201.45. The new term structure is as follow:TermRateDiscount bond price60 days6.80%B30(60)= 1/(1+.068(60/360)) = 0.9888150 days7.05%B30(150)= 1/(1+.0705(150/360)) = 0.9715Find the value of swap 1, 2 and 3.

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